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Unit 2: Vector space

Vector:

$8\hat{i} + 9\hat{j} + 10\hat{k}$, $\begin{bmatrix} 8 \\ 9 \\ 10 \end{bmatrix}$ or $\begin{bmatrix} 8 & 9 & 10 \end{bmatrix}$

Vector in $\mathbb{R}^n$

It is set of n tuples of real nos. It is written as $[a_1, a_2, a_3 \dots a_n]$. It represents a point in n-dimensional space and each element in a tuple is referred as a component

Vector addition

Let $U = [U_1, U_2 \dots U_n]$
$V = [V_1, V_2 \dots V_n]$, then
$W = U + V$
$= (U_1 + V_1, U_2 + V_2 \dots U_n + V_n)$

Scalar multiplication

Let $U = (U_1, U_2, U_3 \dots U_n)$ and k be any scalar then, $k \times U = (kU_1, kU_2, kU_3, kU_4 \dots kU_n)$

Vector space

It is an non-empty set that can be added together and multiplied by scalars such that it satisfies following condition:

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(1) Closure under addition:

let U & V are 2 vectors in a set V such $U+V$ also belongs to V.

(2) Commutative

$(U,V) \in V$, $U+V = V+U \in V$

(3) Associative of addition

let $U, V, W \in V$, then $(U+V)+W = U+(V+W) \in V$

(4) Additive identity (zero vector)

There is vector 0 in V such that $0+U = U$ $\forall$ $U \in V$

(5) Additive inverse

For each U in V there exist a vector $-U$ such that $U+(-U) = 0$.

(6) Closure under SM

$U \in V$, $kU \in V$.

(7) Distribution under $\times$.

$k(U+V) = kU+kV$

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Ex. $\mathbb{R}^2 = \{(x,y) \mid x, y \in \mathbb{R}\}$

Check whether $V = \{(x,y) \mid x, y \in \mathbb{R}\}$

(1) with vector addition $(x_1, y_1) + (x_2, y_2) = (x_1 y_2, x_2 y_1)$

(2) & scalar multiplication $c(x,y) = (cx, cy)$

is a vector space.

$\Rightarrow$ In this set, the commutative property fails because

$$(x_1, y_1) + (x_2, y_2) \neq (x_2, y_2) + (x_1, y_1)$$

$$(x_1 y_2, x_2 y_1) \neq (x_2 y_1, x_1 y_2)$$

NOTE: $\mathbb{R}^n$ is always a Vector space.

Discuss the examples for VS.

1. $A = \{(x,y,z) / x, y, z \in \mathbb{R}\}$

This is $\mathbb{R}^3$. Set of all ordered triples ~~with~~ let $U = (x_1, y_1, z_1)$ & $V = (x_2, y_2, z_2) \in A$. Then $U+V = (x_1 + x_2, y_1 + y_2, z_1 + z_2) \in A$. Let k be any scalar, then $kU = (kx_1, ky_1, kz_1) \in A$.

$\therefore$ Closure under addition & scalar multiplication will hold & also it satisfies all axioms from Real No. property. Hence A is VS.

2. $D = \left\{ \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \mid a, b \in \mathbb{R} \right\}$

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$\Rightarrow$ Let $U = \begin{bmatrix} a_1 & 0 \\ 0 & b_1 \end{bmatrix}$ & $V = \begin{bmatrix} a_2 & 0 \\ 0 & b_2 \end{bmatrix}$ where $U, V \in D$.

$U + V = \begin{bmatrix} a_1 + a_2 & 0 \\ 0 & b_1 + b_2 \end{bmatrix} \in \mathbb{R}$

Let k be any scalar, then $kU = \begin{bmatrix} ka_1 & 0 \\ 0 & kb_1 \end{bmatrix}$

where $a_1, b_1 \in \mathbb{R}$

$\therefore$ It puts closure property under addition and scalar multiplication & also satisfies other axioms of VS. B is VS.

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$\Rightarrow$ Check whether the given set $V = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}$ is VS or not.

- Let $u = (x_1, y_1, z_1) \in V$, $v = (x_2, y_2, z_2) \in V$, $w = (x_3, y_3, z_3) \in V$ and k, c be any scalars.

a) $u \in V$, $v \in V$ then $U + V = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$

$x_1 + x_2, y_1 + y_2, z_1 + z_2$ are real nos.

$u + v \in V$

b) $u \in V$, k any scalar

$ku = (kx_1, ky_1, kz_1)$, $\therefore ku \in V$

c) $u+v = (x_1+x_2, y_1+y_2, z_1+z_2)$
$v+u = (x_2+x_1, y_2+y_1, z_2+z_1)$

$u+v = v+u$
$\therefore v+u \in V$.

d) $(u+v)+w = ((x_1+x_2)+x_3, (y_1+y_2)+y_3, (z_1+z_2)+z_3)$
$u+(v+w) = (x_1+(x_2+x_3), y_1+(y_2+y_3), z_1+(z_2+z_3))$

$(u+v)+w = u+(v+w)$
$\therefore u+(v+w) \in V$

e) let 0 vector $\in V$
0 vector $= (0,0,0)$
$u+0 = u \in V$

f) let $u = (x_1, y_1, z_1) \in V$, the $-u = (-x_1, -y_1, -z_1) \in V$

$u+(-u) = (x_1+(-x_1), y_1+(-y_1), z_1+(-z_1)) = (0,0,0)$
$\in V$.

g) Distributive ~~commutative~~ under SM

let k be any scalar, $u, v \in V$

$k(u+v) = (k(x_1+x_2), k(y_1+y_2), k(z_1+z_2))$
$= (kx_1+kx_2, ky_1+ky_2, kz_1+kz_2)$
$= (kx_1, ky_1, kz_1) + (kx_2, ky_2, kz_2)$
$= k(x_1, y_1, z_1) + k(x_2, y_2, z_2)$
$= ku + kv \in V$

b) (g) under SA

$(k+l)u = (k+l)(x_1, y_1, z_1)$
$= ((k+l)x_1, (k+l)y_1, (k+l)z_1)$
$= (kx_1 + lx_1, ky_1 + ly_1, kz_1 + lz_1)$
$= (kx_1, ky_1, kz_1) + l(x_1, y_1, z_1)$
$= ku + lu \in V$

i)

$(kl)u = kl(x_1, y_1, z_1)$
$= (klx_1, kly_1, klz_1)$
$= (k(lx_1), k(ly_1), k(lz_1))$
$= \cancel{k(lu), k(lu), k(lu)}$

$= k(lu)$

j) $1u = 1(x_1, y_1, z_1)$

[2] $B = \left\{ \begin{bmatrix} a & b \\ c & 1 \end{bmatrix} \middle| \, a, b, c \in R \right\}$

It is not a VS because let $u = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix} \in B$

and $v = \begin{bmatrix} 5 & 6 \\ 7 & 1 \end{bmatrix} \in B$

$u + v = \begin{bmatrix} 7 & 9 \\ 11 & 2 \end{bmatrix} \notin B$

$\therefore$ B isn't VS.

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Subspace

Let V is a vector space & W is a subset of V, then W is a supspace of V if and only if it satisfies following conditions:

- It should contain a $\vec{0}$:
- Closure under addition: If $u, v \in W$, then $u+v \in W$
- — " — SM: If $v \in W$ & k be any scalar, then $kv \in W$

NOTE: To verify that a subset W is a subspace of V.S 'V'

(i) Check if $\vec{0}$ is in W.
(ii) Verify that $u, v \in W$ & $u+v \in W$.
(iii) Verify that $v \in W$ & $kv \in W$.

3/2/26 Check wheter the set is a subspace or not.

1. $B = \{(x, y, z) \mid x^2+y^2+z^2 \le 1\}$ is subspace of $\mathbb{R}^3$ or not

Let $\vec{0}$ belongs to B, then $0^2+0^2+0^2 = 0 \le$

$\therefore \vec{0}$ belongs to B.

2. Closure under addition

Let $U = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ & $V \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, where $U, V \in B$

as $U \Rightarrow 1^2+0^2+0^2 = 1 \le 1$ & $V \Rightarrow 0^2+1^2+0^2 = 1 \le$

but $U+V = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = 1^2+1^2+0^2 = 2 \nleq 1$

$\therefore U+V \notin B$

$\therefore B$ is not a subspace of $\mathbb{R}^3$

2. $C = \{(x,y) \mid xy \leq 1\}$ is a subspace or not

$\Rightarrow$ It is not a subspace because,

let $U = (1,1)$ & $V = (0,1)$

$U = 1 \times 1 = 1 \leq 1 \in C$
$V = 0 \times 1 = 0 \leq 1 \in C$

$U+V = (1,2) = 1 \times 2 = 2 \nleq 1 \notin C$

$\therefore U+V \notin C$

$\therefore C$ is not a subspace.

3. $E = \{(x,y) \mid x^2+y^2 \geq 0\}$ is a subspace of R^2 or not.

$\Rightarrow$ let $(0,0) \in E$ then $0^2+0^2 = 0 \geq 0$,

$\therefore \vec{0} \in E$

Closure under addition

let $U = (x_1, y_1) \in E$ & $V = (x_2, y_2) \in E$

i.e.

$U = (x_1, y_1) = x_1^2 + y_1^2 \geq 0$

$V = (x_2, y_2) = x_2^2 + y_2^2 \geq 0$

then $U + V = (x_1, y_1) + (x_2, y_2)$

$= (x_1 + x_2), (y_1 + y_2)$

$= (x_1 + x_2)^2 + (y_1 + y_2)^2 \geq 0$

because the sum of squares is always non-ve for R

$\therefore U + V \in E$

Closure under scalar multiplication

let c be any scalar and $U = (x, y) \in E$, then,

$cu = c(x_1, y_1)$

$= (cx_1, cy_1)$

$= (cx_1)^2 + (cy_1)^2 \geq 0$

$\therefore cu \in E$

$\therefore E$ is an subspace

4. $W = \{(x, y, z) \mid x \geq 0\}$ is a subspace of R^3

$\Rightarrow$ It is not a subspace because, let $U = (3, -4, 0)$ where $3 \geq 0$, $U \in W$

let $c = -1$ be any scalar then $cu = -1(3,4,0)$
$= (-3,4,0)$, $-3 \ngeq 0$
$\therefore cu \notin W$. It fails under SM
$\therefore W$ is not a sub space.

Theorem

If W_1 & W_2 are subspaces of a vector space V then $W_1 + W_2 =$ ~~by~~ $x+y$, $x \in W_1$ & $y \in W_2$ is a subspace of V.

Since W_1 & W_2 are the subspaces, so $0 \in W_1$, $0 \in W_2$, then $0+0 = 0 \in W_1 + W_2$. Thus $W_1 + W_2$ is a non-empty set.

Closure under addition.

Let $u, v \in W_1 + W_2$

$u = x_1 + y_1$, $v = x_2 + y_2$

For sum, $x_1, x_2 \in W_1$
$y_1, y_2 \in W_2$

then $u+v = (x_1+y_1) + (x_2+y_2)$
$(x_1+x_2) + (y_1+y_2)$

Since W_1 & W_2 are subspaces, $x_1 + x_2 \in W_1$ & $y_1 + y_2 \in W_2$. Hence $u+v \in W_1 + W_2$

Closure under Scalar Multiplication

$u = x_1 + y_1 \in W_1 + W_2$

where $x_1 \in W_1$ & $y_1 \in W_2$

let c be any scalar,

then $cu = c(x_1 + y_1)$
$cu = cx_1 + cy_1$

Since W_1 & W_2 are subspaces,

$cx_1 \in W_1$ & $cy_1 \in W_2$

Hence $cu = cx_1 + cy_1 \in W_1 + W_2$

$\therefore W_1 + W_2$ is a subspace

4/2 Theorem

If G & H are subspaces of V.S V. Prove that $G \cap H$ is also a subspace. Is $G \cup H$ a subspace? Justify with an example.

→ Since G & H are the subspaces,
$0 \in G$ & $0 \in H$
$\therefore 0 \in G \cap H$

So $G \cap H \neq \phi$

Take any $x, y \in G \cap H$, then

$x, y \in G$ & $x, y \in H$ because G & H are subspaces.

Then $x+y \in G$ & $x+y \in H$ ~~Since~~ Hence $x+y \in G \cap H$

Let $x \in G \cap H$, then $x \in G$ & $x \in H$
Let c be any scalar,
Since G & H are subspace, $cx \in G$, $cx \in H$
Then $cx \in G \cap H$

$\therefore G \cap H$ is a subspace.

Let $V = \mathbb{R}^2$; $G = \{(x,0) / x \in \mathbb{R}\}$; $H = \{(y,0) / y \in \mathbb{R}\}$, then take $(1,0) \in G \cup H$ & $(0,1) \in G \cup H$

$(1,0) + (0,1) = (1,1) \notin G \cup H$

Closure under addition is not satisfied.
$\therefore G \cup H$ is not a subspace.

Linear Combination

Let V be a V.S. with the vectors $v_1, v_2, \dots v_n$ then the expression $c_1 v_1 + c_2 v_2 + \dots + c_n v_n$ is called linear combination where $c_1, c_2, \dots, c_n$ are called scalars.

Ex: $2i + 3j + 4k = 2 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 4 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

where 2, 3, 4 are c & $\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix} \& \begin{bmatrix}0\\0\\1\end{bmatrix}$ are

Q. Express the given vector as LC.

1. $V = [5, 8]$ as L.C of $(1,1)$ & $(-1,1)$

$$\begin{bmatrix}5\\8\end{bmatrix} = c_1\begin{bmatrix}1\\1\end{bmatrix} + c_2\begin{bmatrix}-1\\1\end{bmatrix}$$

$$5 = c_1 - c_2$$
$$8 = c_1 + c_2$$

$$13 = 2c_1$$
$$c_1 = \frac{13}{2}$$

$$8 = 6.5 + c_2$$
$$c_2 = 1.5 = \frac{3}{2}$$

$$\therefore \begin{bmatrix}5\\8\end{bmatrix} = \frac{13}{2}\begin{bmatrix}1\\1\end{bmatrix} + \frac{3}{2}\begin{bmatrix}-1\\1\end{bmatrix}$$

2. $V = (1, -2, 5)$ as L.C of $(1,1,1)$ $(1,2,3)$ $(2,1,1)$

$$\begin{bmatrix}1\\-2\\5\end{bmatrix} = c_1\begin{bmatrix}1\\1\\1\end{bmatrix} + c_2\begin{bmatrix}1\\2\\3\end{bmatrix} + c_3\begin{bmatrix}2\\1\\1\end{bmatrix}$$

$c_1 + c_2 + 2c_3 = 1$ — ①
$c_1 + 2c_2 + c_3 = -2$ — ②
$c_1 + 3c_2 + c_3 = 5$

$2c_1 + 2c_2 + \cancel{2c_3} = 2$
$c_1 + 2c_2 + \cancel{c_3} = -2$
$c_1 \quad + 3c_3 = 4$

$\cancel{3c_2 - 2c_3 - 1}$
$c_2 = \cancel{1/3}$

$\cancel{c_1 + c_3}$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 1 & 2 & 1 & -2 \\ 1 & 3 & 1 & 5 \end{array}\right]$$

$R_2 \to R_2 - R_1$
$R_3 \to R_3 - R_1$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & -1 & -3 \\ 0 & 2 & -1 & 4 \end{array}\right]$$

$R_3 \to R_3 - 2R_2$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 1 & 10 \end{array}\right]$$

$c_1 + c_2 + 2c_3 = 1$
$c_2 - c_3 = -3$
$c_3 = 10$

$\Rightarrow c_2 = 7 \qquad \Rightarrow c_1 + 7 + 20 = 1$
$c_1 = -26$

$$\begin{bmatrix} 1 \\ -2 \\ 5 \end{bmatrix} = -26 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + 7 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + 10 \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

3. $U=(2,13,6)$ as L.C. of $(1,5,-1)\ (1,2,1)\ (1,4,3)$

4. $U=(3,7,-4)$ 〃 $(1,2,3)\ (2,3,7)\ (3,5,6)$

5. $U=(2,-5,3)$ 〃 $(1,-3,2)\ (2,-4,-1)\ (1,-5,\ldots)$

3. $\begin{bmatrix}2\\13\\6\end{bmatrix} = c_1\begin{bmatrix}1\\5\\-1\end{bmatrix} + c_2\begin{bmatrix}1\\2\\1\end{bmatrix} + c_3\begin{bmatrix}1\\4\\3\end{bmatrix}$

$c_1+c_2+c_3=2$
$5c_1+2c_2+4c_3=13$
$-c_1+c_2+3c_3=6$

$\left[\begin{array}{ccc:c}1&1&1&2\\5&2&4&13\\-1&1&3&6\end{array}\right]$ $\quad R_2\rightarrow R_2-5R_1$, $\;R_3\rightarrow R_3+R_1$

$\left[\begin{array}{ccc:c}1&1&1&2\\0&-3&-1&3\\0&2&4&8\end{array}\right]$ $\quad R_3\rightarrow 3R_3+2R_2$

$\left[\begin{array}{ccc:c}1&1&1&2\\0&-3&-1&3\\0&0&10&30\end{array}\right] \Rightarrow$ $\begin{aligned}c_1+c_2+c_3&=2\\-3c_2-c_3&=3\\10c_3&=30\end{aligned}$

$c_3=3$
$c_2=-2$
$\underline{\underline{c_1=1}}$

$\therefore \begin{bmatrix}2\\13\\6\end{bmatrix} = 1\begin{bmatrix}1\\5\\-1\end{bmatrix} - 2\begin{bmatrix}1\\2\\1\end{bmatrix} + 3\begin{bmatrix}1\\4\\3\end{bmatrix}$

4. $\begin{bmatrix}3\\7\\-4\end{bmatrix} = c_1\begin{bmatrix}1\\2\\3\end{bmatrix} + c_2\begin{bmatrix}2\\3\\7\end{bmatrix} + c_3\begin{bmatrix}3\\5\\6\end{bmatrix}$

$$\left[\begin{array}{ccc:c} 1 & 2 & 3 & 3 \\ 2 & 3 & 5 & 7 \\ 3 & 7 & 6 & -4 \end{array}\right] \qquad \begin{array}{l} R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 - 3R_1 \end{array}$$

$$\left[\begin{array}{ccc:c} 1 & 2 & 3 & 3 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & -3 & -13 \end{array}\right] \qquad R_3 \to R_3 + R_2$$

$$\left[\begin{array}{ccc:c} 1 & 2 & 3 & 3 \\ 0 & -1 & -1 & 1 \\ 0 & 0 & -4 & -12 \end{array}\right] \qquad \begin{array}{r} c_1 + 2c_2 + 3c_3 = 3 \\ -c_2 - c_3 = 1 \\ -4c_3 = -12 \end{array}$$

$c_3 = 3$; $c_2 = -4$; $c_1 = 2$

$$\therefore \begin{bmatrix} 3 \\ 7 \\ -4 \end{bmatrix} = 2\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - 4\begin{bmatrix} 2 \\ 3 \\ 7 \end{bmatrix} + 3\begin{bmatrix} 3 \\ 5 \\ 6 \end{bmatrix}$$

5 $$\begin{bmatrix} 2 \\ -5 \\ 3 \end{bmatrix} = c_1\begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} + c_2\begin{bmatrix} 2 \\ -4 \\ -1 \end{bmatrix} + c_3\begin{bmatrix} 1 \\ -5 \\ 7 \end{bmatrix}$$

$$\left[\begin{array}{ccc:c} 1 & 2 & 1 & 2 \\ -3 & -4 & -5 & -5 \\ 2 & -1 & 7 & 3 \end{array}\right] \qquad \begin{array}{l} R_2 \to R_2 + 3R_1 \\ R_3 \to R_3 - 2R_1 \end{array}$$

$$\left[\begin{array}{ccc:c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 1 \\ 0 & -5 & 5 & -1 \end{array}\right] \qquad R_3 \to 2R_3 + 5R_2$$

$$\left[\begin{array}{ccc:c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 1 \\ 0 & 0 & 0 & 3 \end{array}\right] \Rightarrow \begin{array}{r} \cancel{c_1 + 2c_2 + c_3 = 2} \\ \cancel{2c_2 - 2c_3 = 1} \end{array}$$

$C_2 = \frac{1+2C_3}{2}$

$C_1 + 1 + 2C_3 + C_3 = 2$

$C_1 = 1 - 3C_3$

$$\begin{bmatrix} 2 \\ -5 \\ 3 \end{bmatrix} = 1-3C_3 \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} + \frac{1+2C_3}{2} \begin{bmatrix} 2 \\ -4 \\ -1 \end{bmatrix} + C_3 \begin{bmatrix} 1 \\ -5 \\ 7 \end{bmatrix}$$

$\rho(A) \neq \rho(A:B)$

$\therefore$ No solution

2 Linear span

Let 'S' be subset of V.S. 'V', then set of all LCs of 'S' is called Linear span of 'S' denoted by $L(S)$ i.e. if $S = V_1, V_2, \ldots, V_n$ then $L(S) = \{C_1V_1, C_2V_2, \ldots, C_nV_n\}$ where $(C_1, C_2 \ldots C_n)$ are scalars.

Ex: let $V = \mathbb{R}^3$

$$L(S) = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

i.e. Any vector $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ in $\mathbb{R}^3$ can be written as L.C of $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = x\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + y\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + z\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Linear Independant

Let 'V' be a V.S with the vectors $(v_1, v_2 \dots v_n)$ then we say that these vectors are L.I. if $c_1v_1 + c_2v_2 + \dots + c_nv_n = 0$ such that $c_1 = c_2 = \dots = c_n = 0$.

Linear Dependent

Let 'V' be a V S with the vectors $(v_1, v_2 \cdots v_n)$ we say that these vectors are L.D if $c_1v_1 + c_2v_2 + \dots + c_nv_n = 0$ such that all $c_i \neq 0$

NOTE: To check wheter the given vectors are L.I / L.D, we follow these methods.

1. Write L.C. of given vectors i.e. $(c_1v_1 + c_2v_2 + \dots + c_nv_n = 0)$, form the matrix of given vectors and reduce it to echelon form. If all the columns have a pivot element, then we say that the given vectors L.I. otherwise they are L.D.
2. We can also check for 2x2 & 3x3 matrix using the determinant. i.e. After forming the matrix of given vectors if the determinant of that matrix $\neq 0$, then we say that it is L.I. and vice versa.

Sums

Q. Show L.I. / L.D.

1. $\{(1,4,5),\ (4,4,8),\ (3,-3,0)\}$

$\Rightarrow c_1V_1 + c_2V_2 + c_3V_3 = 0$

$$c_1\begin{bmatrix}1\\4\\5\end{bmatrix} + c_2\begin{bmatrix}4\\4\\8\end{bmatrix} + c_3\begin{bmatrix}3\\-3\\0\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$$

$$\left[\begin{array}{ccc:c}1 & 4 & 3 & 0\\4 & 4 & -3 & 0\\5 & 8 & 0 & 0\end{array}\right] \quad \begin{array}{l}R_2 \to R_2 - 4R_1\\ R_3 \to R_3 - 5R_1\end{array}$$

$$\left[\begin{array}{ccc:c}1 & 4 & 3 & 0\\0 & -12 & -15 & 0\\0 & -12 & -15 & 0\end{array}\right] \quad R_3 \to R_3 - R_2$$

$$\left[\begin{array}{ccc:c}1 & 4 & 3 & 0\\0 & -12 & -15 & 0\\0 & 0 & 0 & 0\end{array}\right]$$

Since there are only 2 pivot elements

$\rho(A:B) \neq n$

$\therefore$ This is L.D.

2. $\{(1,0,-1,2),\ (4,2,0,-1),\ (6,4,-2,3)\}$

$$c_1V_1 + c_2V_2 + c_3V_3 = 0$$

$$c_1\begin{bmatrix}1\\0\\-1\\2\end{bmatrix} + c_2\begin{bmatrix}4\\2\\0\\-1\end{bmatrix} + c_3\begin{bmatrix}6\\4\\-2\\3\end{bmatrix} = 0$$

$$\left[\begin{array}{ccc:c}1&4&6&0\\0&2&4&0\\-1&0&-2&0\\2&-1&3&0\end{array}\right] \qquad \begin{array}{l}R_3 \to R_3 + R_1\\ R_4 \to R_4 - 2R_1\end{array}$$

$$\left[\begin{array}{ccc:c}1&4&6&0\\0&2&4&0\\0&4&4&0\\0&-9&-9&0\end{array}\right] \qquad \begin{array}{l}R_3 \to R_3 - 2R_2\\ R_4 \to 2R_4 + 9R_2\end{array}$$

$$\left[\begin{array}{ccc:c}1&4&6&0\\0&2&4&0\\0&0&-4&0\\0&0&18&0\end{array}\right] \qquad \begin{array}{l}R_4 \to \cancel{2R_4 + 3R_3}\\ \quad\; 2R_4 + 9R_3\end{array}$$

$$\left[\begin{array}{ccc:c}1&4&6&0\\0&2&4&0\\0&0&-4&0\\0&0&0&0\end{array}\right] \qquad \begin{array}{l}\cancel{\rho(A:B) = m}\\ \rho(A:B) = n\end{array}$$

3. $\{(1,2,-1)\ (1,-2,1)\ (-3,2,-1)\}$

$$c_1V_1 + c_2V_2 + c_3V_3 = 0$$

$$c_1\begin{bmatrix}1\\2\\-1\end{bmatrix} + c_2\begin{bmatrix}1\\-2\\1\end{bmatrix} + c_3\begin{bmatrix}-3\\2\\-1\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$$

$$\left[\begin{array}{ccc:c} 1 & 1 & -3 & 0 \\ 2 & -2 & 2 & 0 \\ -3 & 1 & -1 & 0 \end{array}\right] \quad \begin{array}{l} R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 + 3R_1 \end{array}$$

$$\left[\begin{array}{ccc:c} 1 & 1 & -3 & 0 \\ 0 & -4 & 8 & 0 \\ 0 & 2 & -4 & 0 \end{array}\right] \quad R_3 \to 2R_3 + R_2$$

$$\left[\begin{array}{ccc:c} 1 & 1 & -3 & 0 \\ 0 & -4 & 8 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad \begin{array}{l} c_1 + c_2 - 3c_3 = 0 \\ -4c_2 + 8c_3 = 0 \end{array}$$

$\rho(A) \neq n$

$\therefore$ L.D

4. $\{(1,0,-1)\ (2,1,3)\ (4,2,6)\}$

$c_1V_1 + c_2V_2 + c_3V_3 = 0$

$$c_1\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + c_2\begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} + c_3\begin{bmatrix} 4 \\ 2 \\ 6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{ccc:c} 1 & 2 & 4 & 0 \\ 0 & 1 & 2 & 0 \\ -1 & 3 & 6 & 0 \end{array}\right] \quad R_3 \to R_3 + R_1$$

$$\left[\begin{array}{ccc:c} 1 & 2 & 4 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 5 & 10 & 0 \end{array}\right] \quad R_3 \to R_3 - 5R_2$$

$$\left[\begin{array}{ccc:c} 1 & 2 & 4 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad \begin{array}{l} c_1 + 2c_2 + 4c_3 = 0 \\ c_2 + 2c_3 = 0 \end{array}$$

$\therefore$ L.D

5. $\{(1,0,-1)\ (2,1,3)\ (4,2,0)\}$

$$c_1 V_1 + c_2 V_2 + c_3 V_3 = 0$$

$$c_1 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} + c_3 \begin{bmatrix} 4 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{ccc:c} 1 & 2 & 4 & 0 \\ 0 & 1 & 2 & 0 \\ -1 & 3 & 0 & 0 \end{array}\right] \qquad R_3 \to R_3 + R_1$$

$$\left[\begin{array}{ccc:c} 1 & 2 & 4 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 5 & 4 & 0 \end{array}\right] \qquad R_3 \to R_3 - 5R_2$$

$$\left[\begin{array}{ccc:c} 1 & 2 & 4 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & -6 & 0 \end{array}\right] \qquad \begin{aligned} c_1 + 2c_2 + 4c_3 &= 0 \\ c_2 + 2c_3 &= 0 \\ -6c_3 &= 0 \end{aligned}$$

$\Rightarrow c_2 = -2c_3$ $\qquad \therefore$ L.I

$\Rightarrow c_1 - 4c_3 + 4c_3 = 0 \quad \Rightarrow c_1 = 0,\ c_2 = 0,\ c_3 = 0$

$\{(1,-1,-2)\ (5,-4,-7)\ (-3,1,0)\}$

$$c_1 V_1 + c_2 V_2 + c_3 V_3 = 0$$

$$c_1 \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix} + c_2 \begin{bmatrix} 5 \\ -4 \\ -7 \end{bmatrix} + c_3 \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{ccc:c} 1 & 5 & -3 & 0 \\ -1 & -4 & 1 & 0 \\ -2 & -7 & 0 & 0 \end{array}\right] \quad \begin{array}{l} R_2 \to R_2 + R_1 \\ R_3 \to R_3 + 2R_1 \end{array}$$

$$\left[\begin{array}{ccc:c} 1 & 5 & -3 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 3 & -6 & 0 \end{array}\right] \quad R_3 \to R_3 - 3R_2$$

$$\left[\begin{array}{ccc:c} 1 & 5 & -3 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad \rho(A) \neq n$$

$c_1 + 5c_2 - 3c_3 = 0$

$c_2 - 2c_3 = 0$

$\therefore$ L.D $\quad \boxed{c_1 = -7c_3} \quad \boxed{c_2 = 2c_3}$

6/1/26 **Basis of Vector Space**

A subset 'S' of V.S 'V' is called Basis if 'S' is L.I & 'S' will span entire 'V'.

(Let 'S' = $v_1, v_2, \dots, v_n$ of Vectors is a basis of V if $u \in$ 'V' can be written as L.C. of Basis vector. i.e. $u = c_1 v_1 + c_2 v_2 \dots, c_n v_n$).

Dimension of Vector Space

No. of Vectors present in basis of V.S 'V' is dimension of V.S. Denoted by dim(V).

Ex. $V = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ forms basis of $\mathbb{R}^3$

dim (V) = 3

$V = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$ forms basis for 2×2 sq. matrix

$\dim(V) = \underline{\underline{4}}$

<u>NOTE</u>: let 'V' be a V.S. of finite dim n. Then any $n+1$ or more vectors in V are L.D.

If V.S. 'V' has basis $= n$ Then each of n should contain n vectors.

10/2 <u>Sums</u>

1. $\{(1,3,1)\ (0,2,0)\ (0,0,7)\}$ is a basis of $\mathbb{R}^3$

$\Rightarrow$ The given vectors should be L.I & these vectors will span entire $\mathbb{R}^3$. To check L.I or L.D.

$c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$

$$c_1 \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 \\ 0 \\ 7 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$[A:B] = \left[\begin{array}{ccc:c} 1 & 0 & 0 & 0 \\ 3 & 2 & 0 & 0 \\ 1 & 0 & 7 & 0 \end{array}\right] \quad \begin{array}{l} R_2 \to R_2 - 3R_1 \\ R_3 \to R_3 - R_1 \end{array}$$

$$\Rightarrow \left[\begin{array}{ccc:c} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 7 & 0 \end{array}\right] \quad c_1 = 0\ ;\ 2c_2 = 0\ ;\ 7c_3 = 0$$

$c_2 = 0 \qquad c_3 =$

$\therefore$ Vector [illegible] L.I.

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Let $u = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ be any vector in $\mathbb{R}^3$, $x, y, z \in$ a, b, c be any scalars, then

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = aV_1 + bV_2 + cV_3$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = a\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} + b\begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} + c\begin{bmatrix} 0 \\ 0 \\ 7 \end{bmatrix}$$

$a = x$ — ①
$3a + 2b = y$ — ②
$a + 7c = z$ — ③

From ① $a = x$
From ② $b = \frac{y - 3x}{2}$
From ③ $c = \frac{z - x}{7}$

Since $x, y, z \in \mathbb{R}$, we can determine a, b, c
Given vector span $\mathbb{R}^3$
Given vectors are basis of $\mathbb{R}^3$

2. $\{(1, -1, 1)\ (1, 0, 0)\ (1, 1, 1)\ (1, 2, 4)\}$ is a basis of $\mathbb{R}^3$

$\Rightarrow$ $c_1V_1 + c_2V_2 + c_3V_3 + c_4V_4 = 0$

$$c_1\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} + c_2\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + c_3\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + c_4\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$[A:B] = \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ -1 & 0 & 1 & 2 & 0 \\ 1 & 0 & 1 & 4 & 0 \end{array}\right] \quad \begin{array}{l} R_2 \to R_2 + R_1 \\ R_3 \to R_3 - R_1 \end{array}$$

$$A \Rightarrow \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & -1 & 0 & 3 & 0 \end{array}\right] \quad R_3 \to R_3 + R_2$$

$$\Rightarrow \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & 2 & 6 & 0 \end{array}\right]$$

$$c_1 + c_2 + c_3 + c_4 = 0$$
$$c_2 + 2c_3 + 3c_4 = 0$$
$$2c_3 + 6c_4 = 0$$

No. of pivot elements $\neq$ No. of columns

$\therefore$ Vector is L.D.

3. $\{(1,0,0)\ (2,3,0)\ (4,5,6)\}$ from the basis of $\mathbb{R}^3$

$\Rightarrow$ $c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$

$$c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} + c_4 \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = 0$$

$$[A:B] = \left[\begin{array}{ccc|c} 1 & 2 & 4 & 0 \\ 0 & 3 & 5 & 0 \\ 0 & 0 & 6 & 0 \end{array}\right] \Rightarrow \begin{array}{r} c_1 + 2c_2 + 4c_3 = 0 \\ 3c_2 + 5c_3 = 0 \\ 6c_3 = 0 \end{array}$$

$c_1 = 0$; $c_2 = 0$; $c_3 = 0$

$\therefore$ Vector is L.I.

let $u = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ be any vector in $\mathbb{R}^3$, x, y, z be any scalar. then

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = aV_1 + bV_2 + cV_3$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = a\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + b\begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} + c\begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

$a + 2b + 4c = x$ — ①

$3b + 5c = y$ — ②

$6c = z$ — ③

$\frac{z}{6} = c$

$b = \frac{6y - 5z}{18}$ $\qquad b = \frac{y - 5c}{3}$ $\qquad \frac{y - \frac{5z}{6}}{3}$

$\frac{6y - 5z}{18}$ $\qquad$ 11/2/26

$a = \frac{-6y + 5z}{9} - \frac{2z}{3} + x$

$a = \frac{-6y - z + 9x}{9}$

Since $x, y, z \in \mathbb{R}$, we can determine a, b, c

Given vector is basis of $\mathbb{R}^3$.

4. $\{(1,-2,3)\,(-2,7,-9)\}$ forms basis of $\mathbb{R}^2$

$\Rightarrow c_1 V_1 + c_2 V_2 = 0$

$$c_1\begin{bmatrix}1\\-2\\3\end{bmatrix} + c_2\begin{bmatrix}-2\\7\\-9\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$$

$$[A:B] = \begin{bmatrix}1 & -2 & : & 0\\ -2 & 7 & : & 0\\ 3 & -9 & : & 0\end{bmatrix} \quad \begin{matrix}R_2 \to R_2 + 2R_1\\ R_3 \to R_3 - 3R_1\end{matrix}$$

$$\Rightarrow \begin{bmatrix}1 & -2 & : & 0\\ 0 & 3 & : & 0\\ 0 & -3 & : & 0\end{bmatrix} \quad R_3 \to R_3 + R_2$$

$$\Rightarrow \begin{bmatrix}1 & -2 & : & 0\\ 0 & 3 & : & 0\\ 0 & 0 & : & 0\end{bmatrix}$$

No. of pivot elements $\neq$ No. of columns

$\therefore$ Vector is L.D.

11/2/26 **Basis & Dimension of Subspace.**

Let W be a subspace of V.S 'V' is set of vectors $v_1, v_2, \ldots, v_n$ are subset of W is called basis of W if

① These vectors are L.I.

② These vectors span W i.e. $W = \text{span}\{v_1, \ldots, v_n\}$

The no. of vectors in W is called dimension of W.

Sums

Q. Find basis & dimension of given subspace

1. $W = \mathbb{R}^3$ spanned by $\{(-1,2,2)(2,2,-1)(3,0,3)(2,-1,2)\}$

$$\Rightarrow \begin{bmatrix} -1 & 2 & 3 & 2 \\ 2 & 2 & 0 & -1 \\ 2 & -1 & 3 & 2 \end{bmatrix} \quad \begin{matrix} R_2 \to R_2 + 2R_1 \\ R_3 \to R_3 + 2R_1 \end{matrix}$$

$$\begin{bmatrix} -1 & 2 & 3 & 2 \\ 0 & 6 & 6 & 3 \\ 0 & 3 & 9 & 6 \end{bmatrix} \quad R_3 \to 2R_3 - R_2$$

$$\begin{bmatrix} -1 & 2 & 3 & 2 \\ 0 & 6 & 6 & 3 \\ 0 & 0 & 12 & 9 \end{bmatrix}$$

Basis of $W = \{(-1,2,2)(2,2,-1)(3,0,3)\}$

2. Determine whether $\{(1,1,1,1)(1,2,3,2)(2,5,6,4)(2,6,8,5)\}$ is basis or not. If not find basis & dimension of subspace spanned by these vectors.

$$\Rightarrow \begin{bmatrix} 1 & 1 & 2 & 2 \\ 1 & 2 & 5 & 6 \\ 1 & 3 & 6 & 8 \\ 1 & 2 & 4 & 5 \end{bmatrix} \quad \begin{matrix} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1 \\ R_4 \to R_4 - R_1 \end{matrix}$$

$$\begin{bmatrix} 1 & 1 & 2 & 2 \\ 0 & 1 & 3 & 4 \\ 0 & 2 & 4 & 6 \\ 0 & 1 & 2 & 3 \end{bmatrix} \quad \begin{array}{l} R_3 \to R_3 - 2R_2 \\ R_4 \to R_4 - R_2 \end{array}$$

$$\begin{bmatrix} 1 & 1 & 2 & 2 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & -1 & -1 \end{bmatrix} \quad R_4 \to 2R_4 - R_3$$

$$\begin{bmatrix} 1 & 1 & 2 & 2 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The given vectors are L.D. which will not form the basis of the Subspace of $\mathbb{R}^4$. The vectors which form basis of $\mathbb{R}^4$ are $\{(1,1,1,1)\ (1,2,3,2)\ (2,5,6,4)\}$

3. $\{(1,-2,5,-3)\ (3,8,-3,-5)\ (2,3,1,-4)\}$

① Find subspace of w

② Extend the basis of $\mathbb{R}^4$

$$\Rightarrow \begin{bmatrix} 1 & 3 & 2 \\ -2 & 8 & 3 \\ 5 & -3 & 1 \\ -3 & -5 & -4 \end{bmatrix} \quad \begin{array}{l} R_2 \to R_2 + 2R_1 \\ R_3 \to R_3 - 5R_1 \\ R_4 \to R_4 + 3R_1 \end{array}$$

$$\begin{bmatrix} 1 & 3 & 2 \\ 0 & 14 & 7 \\ 0 & -18 & -9 \\ 0 & 4 & 2 \end{bmatrix} \quad \begin{matrix} R_3 \to 14R_3 + 18R_2 \\ R_4 \to 7R_4 - 2R_2 \end{matrix}$$

$$\begin{bmatrix} 1 & 3 & 2 \\ 0 & 14 & 7 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

(2) To extend the basis of W to $\mathbb{R}^4$, add any 2 vectors which are L.I.

$\therefore$ Basis of $\mathbb{R}^4$ are $\{(1,-2,5,-3)(3,8,-3,-5)$
$(0,0,1,0)(0,0,0,1)\}$

12/2 <u>Coordinate vectors</u>

Let V be V.S. & $S = \{v_1, v_2, \dots v_n\}$ be the basis of V, then for each vector $u \in V$, there exist a unique set of scalars $C = \{c_1, c_2, \dots, c_n\}$ such that $u = c_1v_1 + c_2v_2 + \dots c_nv_n$. C is called coordinate vector of u relative to S. Denoted $[u]_S = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$

<u>Sums</u>

Q. Find coordinate vectors.

1. $B = \{(1,-3)\ (2,-5)\}$, $X = (-2,1)$

$$\begin{bmatrix} -2 \\ 1 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ -3 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ -5 \end{bmatrix}$$

$c_1 + 2c_2 = -2$ — ①
$-3c_1 - 5c_2 = 1$ — ②

$\times 3$ to ①

$$\begin{aligned} 3c_1 + 6c_2 &= -6 \\ -3c_1 - 5c_2 &= 1 \\ \hline c_2 &= -5 \end{aligned}$$

$c_1 - 10 = -2$
$c_1 = 8$

$$[X]_B = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 8 \\ -5 \end{bmatrix}$$

2. $B = \{(1,-1,-3)(-3,4,9)(2,-2,4)\}$
$X = \{(8,-9,6)\}$

$$\begin{bmatrix} 8 \\ -9 \\ 6 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ -1 \\ -3 \end{bmatrix} + c_2 \begin{bmatrix} -3 \\ 4 \\ 9 \end{bmatrix} + c_3 \begin{bmatrix} 2 \\ -2 \\ 4 \end{bmatrix}$$

$c_1 - 3c_2 + 2c_3 = 8$
$-c_1 + 4c_2 - 2c_3 = -9$
$-3c_1 + 9c_2 + 4c_3 = 6$

$$\left[\begin{array}{ccc:c} 1 & -3 & 2 & 8 \\ -1 & 4 & -2 & -9 \\ -3 & 9 & 4 & 6 \end{array}\right] \quad \begin{array}{l} R_2 \to R_2 + R_1 \\ R_3 \to R_3 + 3R_1 \end{array}$$

$$\left[\begin{array}{ccc:c} 1 & -3 & 2 & 8 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 10 & 30 \end{array}\right]$$

$c_1 - 3c_2 + 2c_3 = 8$

$c_2 = -1$

$10c_3 = 30$

$c_3 = 3$

$c_1 + 3 + 6 = 8$

$c_1 = -1$

$$\begin{bmatrix} X \end{bmatrix}_B = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 3 \end{bmatrix}$$

3. $B = \{(1,1,+1)\,(1,1,0)\,(1,0,0)\}$

$X = \{(a,b,c)\}$

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \\ +1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{ccc:c} 1 & 1 & 1 & a \\ 1 & 1 & 0 & b \\ +1 & 0 & 0 & c \end{array}\right] \quad \begin{array}{l} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1 \end{array}$$

$$\left[\begin{array}{ccc:c} 1 & 1 & 1 & a \\ 0 & 0 & -1 & b-a \\ 0 & -1 & -1 & c-a \end{array}\right] \quad \begin{array}{l} \cancel{R_3 \to R_3 + R_2} \\ R_3 \Leftrightarrow R_2 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 0 & -1 & -1 & c-a \\ 0 & 0 & -1 & b-a \end{array}\right]$$

$$c_1 + c_2 + c_3 = a$$
$$-c_2 - c_3 = c - a$$
$$-c_3 = b - a$$

$$\Rightarrow c_3 = a - b$$

$$-c_2 + b - a = c - a$$
$$-c_2 = c - b$$
$$c_2 = b - c$$

$$c_1 + a - b + b - c = a$$
$$c_1 = c$$

$$[x]_B = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} c \\ b-c \\ a-b \end{bmatrix}$$

4. $\begin{bmatrix} 5 \\ 3 \end{bmatrix} = c_1 \begin{bmatrix} 3 \\ -5 \end{bmatrix} + c_2 \begin{bmatrix} -4 \\ 6 \end{bmatrix}$

$$3c_1 - 4c_2 = 5 \quad \text{—— (1)}$$
$$-5c_1 + 6c_2 = 3 \quad \text{—— (2)}$$

$\times 3$ (1) $\quad \times 2$ (2)

$$9c_1 - 12c_2 = 15$$
$$-10c_1 + 12c_2 = 6$$
$$-c_1 = 21$$

$\Rightarrow c_1 = -21$

$c_2 = 87$

$$[x]_B = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} -21 \\ -17 \end{bmatrix}$$

Rowspace & Column Space

Consider a matrix A of order $m \times n$.

$$A = \begin{array}{c} \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{array}{l} - R_1 \\ - R_2 \\ \\ - R_m \end{array} \\ \begin{array}{cccc} C_1 & C_2 & & C_n \end{array} \end{array}$$

Rowspace of matrix A is set of all L.C of its Row vectors. Then Rowspace $R(A)$ = Span of Row vectors of A i.e. $= \{(\alpha_1 R_1 + \alpha_2 R_2 + \dots \alpha_n R_m)\}$ where $\alpha_1, \alpha_2, \dots \alpha_m$ are scalars.

Columnspace of matrix A is set of all L.C. of its column vectors.